Reviewer 1

We thank reviewer #1 for their very thorough reading and feedback here, it has really helped us to improve the manuscript. Our responses are as follows:

*Abstract

Line 7, "Nostoc-feather-moss": The second hyphen here ("feather-moss") is unnecessary and makes this a bit confusing for readers who are not familiar with mosses. I advise you to remove the hyphen to improve the readability of the paper.

- We have removed the second hyphen in all cases where it was used.

*Introduction

Line 17 and elsewhere, "nifH": Don't forget to italicize the gene names.

- Done.

Line 46, "Sphagnum"; Lines 54, 56, 58 and others throughout the manuscript — "Nostoc": Please don't forget to italicize genus and species names either.

- we have searched all instances of scientific species names and corrected non-italicized names.
- in the process we noticed several instances where family names had been italicized, and we have reverted these to unformatted text.

Lines 66-67, "fixed organic nitrogen in the form of fixed, organic nitrogen": This sentence is redundant, so could you please rephrase it?

- this sentence has been changed (previous lines 66-70, current lines 76-78):

From:

"Nutrient exchanges are hypothesized, involving a transfer of fixed organic nitrogen in the form of fixed, organic nitrogen from heterocystous Nostoc colonies to the host plant (Arróniz-Crespo et al., 2022; Bay et al., 2013), perhaps in exchange for organic carbon- and sulfur-containing materials from the moss host, though the latter has only recently been directly reported (Stuart et al., 2020)."

To:

"Nutrient exchanges are hypothesized, in the form of fixed, organic nitrogen from heterocystous *Nostoc* colonies to the host plant (Arróniz-Crespo et al., 2022; Bay et al., 2013), perhaps in exchange for organic carbon- and sulfur-containing materials from the moss host, though the latter has only recently been directly reported (Stuart et al., 2020)."

Line 74: This figure legend is super short and not very informative, so there's room to improve it. What do "o", "m" and "u" mean? What are the circles representing? Where exactly are the Nostoc colonies? Does (c) mean ©?

- We have attempted to provide more detail in the caption:

From:

"**Figure 1**: Hylocomium splendens with colonies of Nostoc ((c) J. Motzkus 2024)"

To:

"**Figure 1**: *Hylocomium splendens* moss with colonies of *Nostoc* cyanobacteria on leaflets (right, lower) (© J. Motzkus 2024). Plants were sectioned into three zones, from the top (labeled "oben"/"o"), middle (m) and bottom ("unten"/"u") of caulids (Fig. 1), for comparison of relative abundances of *nifH* abundances among different positions."

*Methods

Line 108, "They were determined using standard literature": It's a bit unclear to me what you mean by determined and standard literature. Did you mean that the mosses were taxonomically identified using this reference work?

- This is indeed what was meant. We also found an error in the citation in this sentence. We have corrected the citation, and attempted to clarify (previous line 108, current line 168).

From:

"They were determined using standard literature (Frahm, Frei , Lüth 2011)."

To:

"Mosses were taxonomically identified according to Frey et al. (2021)"

Line 117: "Bei" is not part of the location name, right? Can it be translated to "near"? Also, how many samples of each species were taken? Were there any replicates?

- "Bei" can indeed be translated to "near", and this has been done.

- the sampling for *Nostoc*-like organisms was not systematic or quantitative. As we mention, the field stage of this study "piggy-backed" on broader moss diversity surveys of sites. These surveys were conservation-focused, intuitive-controlled Bryophyte surveys of various sites throughout our region, conducted by M. Feulner, in contract with local land-management agencies.

As such, the definition of "replicate" is a little difficult to give here. Many dozens of moss gametophytes were examined for the presence of *Nostoc*-like colonies, but the observation effort varied, from quick examination with a hand lens for first indications of blue-green discoloration, to full microscopic examination with fluorescence microscopy. Some samples that were collected from the field because they presented some indication that they might host *Nostoc* due to discoloration or visible colonies, while other mosses were only serendipitously noticed to host *Nostoc*-like colonies during the process of taxonomically identifying mosses using microscopic traits. We assume, based on the frequency with which cyanobacteria are detected in mosses with molecular methods, that our microscopy and culture-based pipeline failed to detect most of the *Nostoc*-like organisms actually present in these mosses, and that our efforts should not be interpreted quantitatively. Rather, we intended the pipeline to provide clear evidence for the presence of this symbiosis in our region with photography and identified cultures. If we report a statistic for observation effort, we will likely give the impression of extreme rarity in these *Nostoc*-like organisms, but it is our opinion that this would be uninformative or even misleading.

Lines 118, 119, 128: The formatting of the titles and subtitles of the these subsections made the structure of the methods quite confusing. It looks like Microbiology is another section at the same

level as Methods instead of a subsection. Please review this to make sure it properly follows Oecologia's article format.

- we agree and have removed the “Microbiology” header and the “Overview” section (original lines 118-127). These were intended to help give the reader an overview of the methods used, but they are confusing. Instead we have added a supplementary graphic with a schematic of laboratory (Supp. fig. 1) and bioinformatic methods (Supp. fig. 2). The following sentence has been added after the “field” section in Materials and Methods (current line 185):

  “Graphical overviews of laboratory and bioinformatics are available (supp. fig. 1 and 2).”

  We hope this helps to clarify both with the issue raised here by Reviewer 1 and with other related questions/issues raised below.

Lines 120-122: Which equipment did you use for this?

- original lines 118-127 are now deleted, as discussed above. We made the following addition (original lines 130-131, current lines 187-190):

  From:
  “Preliminary surveys for the presence of Nostoc were conducted with standard light microscopy, first using binocular stereoscopes (Wild HeerBrug M-series) and later using Zeiss compound microscopes.”

  To:
  “Preliminary surveys for the presence of *Nostoc* were conducted with standard light microscopy, first using binocular stereoscope (MDG-17, Wild HeerBrugg, Switzerland), followed by compound light microscopy (Leitz HM-Lux 3 and Revolve 2, ECHO, San Diego, USA).”

Line 123, "amplification of NifH and 16S rRNA marker genes": A bit of a small detail, but NifH refers to the protein, not to the marker gene.

- original lines 118-127 are now deleted, as discussed above. “NifH” and “nifH” have been corrected elsewhere to “*nifH”* as suggested above by reviewer 1.

Lines 122-124: How was PCR performed?

- original lines 118-127 are now deleted, as discussed above. Details of PCR are given in the section “PCR” (original line 192, current lines 265-314). Given this and Reviewer 1’s additional comment below, we realized that our description of PCRs in this section was very confusing. As such we have reworked the PCR methods section. We hope this has clarified the methods.

Lines 124-125: How was DNA extraction performed?

- original lines 118-127 are now deleted, as discussed above. Details of DNA extraction are given in the section “DNA extraction” (original line 179, current lines 265-314). DNA extraction was not necessary for 16S rRNA and *nifH* barcoding, as these data were generated via direct colony PCRs.

Lines 125-127: How was surface sterilization performed?

- original lines 118-127 are now deleted, as discussed above. Details of surface sterilization were given in the section “Surface sterilization” (original line 143, current lines 211-221). The term “surface sterilization” has been replaced by to “Endophyte survey”, to hopefully address Reviewer 1’s comments below.

Lines 130-131: Please add manufacturer and model information for the microscopes used.

- addressed above.

Lines 131-133: How was DNA purification and PCR amplification performed?

- Details of DNA extraction are given in the section “DNA extraction” (original line 179, current lines 246-263). To help clarify which samples were subject to DNA extraction, we changed the following:

From (previous lines131-133):

For comparison, some specimens which were surveyed for Nostoc colonies using light microscope and that were found not to host visible colonies were then subjected to DNA purification and to whole-community PCR amplification of nifH genes in PCR tests.

To (current lines 190-192):

For comparison, some specimens which were surveyed for *Nostoc*-like colonies using light microscopy and which were found not to host visible colonies were then subjected to survey for endophytic cyanobacteria (whole-community DNA extraction and PCR, see next)..

Lines 133-135, "Sectioning was done using sections of one centimeter from the top (o), middle (m) and bottom (u) of caulids": How did you do the sections? Did you just use e.g. a scalpel or was another type of instrument necessary?

- we have added the following language to clarify (original line 136, current lines 195-196):

“Sectioning was done with a scalpel that was flame-sterilized over a Bunsen burner, before and after cuts.”

Lines 140-141, "eukaryotic microalgae": There are no prokaryotic microalgae, so this term is redundant. Microalgae is sufficient here.

- here we are attempting to distinguish the target organisms (Cyanobacteria) from any microscopic organisms with plastids. Cyanobacteria were historically referred to as “blue-green algae”, and are even still occasionally referred to as algae by some authorities (see for instance the title and specifically section of 1.2 of Büdel and Friedl’s 2024 text, *Biology of Algae, Lichens, and Bryophytes*). Other papers on moss-cyanobacteria symbioses also occasionally refer to *Nostoc*-like organisms as algae (see for example Bragina et al. (2012)). Thus if we simply use the term “microalgae” some readers may become confused. The term “green algae” is sometimes used for eukaryotic phototrophic oxygenic microorganisms, but this term can also include macro-algae. We considered the use of the term “micro-green algae”, but we have not seen this termed used before. If the reviewers have an alternative collective term for microscopic Chlorophyta and Streptophyta besides “microalgae” then we are happy to use it to here to clarify. We acknowledge that the strict definition of microalgae does not include Cyanobacteria but we feel that keeping this relatively minor addition (one word) can help to reduce confusion with non-expert readers.

Lines 139-152-142: This long paragraph seems more appropriate for the Introduction or Discussion sections.

- we agree that this paragraph is too lengthy. The level of detail was due to our initial struggles with visualizing these cyanobacteria. The standard sources (e.g. DeLuca et al. (2002) or Ininbergs et al. (2011)) under-describe their methods on this and we had poor luck with their recommended method of UV-range excitation. So we thought our methods with fluorescence microscopy might be useful to other researchers who might encounter similar issues.

- We shortened this paragraph. We would like to keep it in methods. This paragraph now reads as follows (current lines 199-207):

> "Moss tissues and bacterial cultures were observed under an epifluorescence microscope (Revolve 2, ECHO, San Diego, USA), using Revolve Software 4.1.0, app version v7.1.0. With epifluorescence microscopy, cyanobacteria can be easily more discerned from eukaryotic microalgae and plant cells (Grigoryeva, 2020; Wojtasiewicz & Stoń-Egiert, 2016). "Cy5" filter setup (CHROMA 49009, ex: 610-650 nm, em: 660-735 nm), can be used to excite C-phycocyanins (Poniedziałek et al., 2017). Phycoerythrins can be excited using "TXRED" filter settings (CHROMA 49008, ex: 540-580, em: 590-665 nm). Algal and plant cells were included in image overlays, using the "FITC/Cy2" filter set (CHROMA 49002, ex: 450-490 nm, em: 500-550 nm) (Donaldson, 2020; Tobimatsu et al., 2013)."

Line 144-145, "To test the hypothesis that cyanobacteria inhabit endophytic compartments of the host mosses, some *Hylocomium* moss samples were surface sterilized" — It would be easier to follow the reasoning behind your methods if you discussed these and other hypotheses that guided the design of your study in the introduction more explicitly. But, as far as I know, feather mosses like *Hylocomium splendens* do not have proper endophytic compartments due to their simple anatomy. As non-vascular plants, they don't have xylems or phloems, and their organs are quite basic, with tissues composed of a single cell layer. This is different than the case of Sphagnum, for example, which has empty cells filled with water (hyalocytes) that can be colonized by microbes and could indeed be considered endophytic compartments. So would you expect to find bacteria inside stems or leaves, for example? It doesn't seem like there'd be room for them.

- we agree with reviewer 1 that other bryophytes-nostocacean endophytic symbioses (*Sphagnum-,* liverwort-, and hornwort-associated *Nostoc*-like organisms) generally occur in specialized compartments, and we have added this information to our discussion:

From (original lines 487-489):

> Endophytic colonization by *Nostoc* and other cyanobacteria is well documented in other plant- and algal-symbiotic systems, including in hornwort and liverwort hosts (reviewed by Adams and Duggan (2008)).

To (current lines 621-624):

> Endophytic colonization by *Nostoc* and other cyanobacteria is well documented in other plant- and algal-symbiotic systems, including in Sphagnum spp., and hornwort and liverwort hosts (reviewed by Adams and Duggan (2008)), albeit in protective plant compartments such as hyalocytes, auricles, or slime-cavities.

- we agree with reviewer 1 that moss leaflets are only one cell thick in cross-section, so leaflets are not likely to harbor numerous endophytes. We mentioned this in the manuscript (original line 480-481, current line 597). And indeed, we observe possible colonization of the caulids with our microscopy, following surface sterilization, not in the leaflets.

- however, we respectfully disagree with Reviewer 1's argument concerning the lack of room for endophytic microorganisms in non-*Sphagnum* moss tissues. There are numerous studies of putatively endophytic fungi and bacteria in mosses. We cite one example each for bacterial and fungal endophytes of mosses (original line 482, current line 614). The bacterial study, Tveit et al. (2020) documents endophytes from both *Sphagnum* and "brown mosses" (Amblystegiaceae). Granted that as pointed out by Reviewer 1, most of these studies are by subtraction, in that they surface sterilize healthy mosses, then culture out bacteria and fungi, and thereby conclude that these organisms must have originated from internal sites in the moss tissue (i.e. they are endophytes) without ever locating the organisms on the plant to confirm that they are some sort of endophyte. Direct imaging of endophytes in mosses is not as common as with vascular plants, except of course for *Sphagnum*.

  Because of the lack of studies that include images of endophytes in mosses, we must look to what is known for vascular plants, in which endophytes are frequently directly imaged. In their case, endophytic colonization is not necessarily limited to – or even predominated by – spaces like vascular tissues or specialized compartments. To give some well-known examples: perhaps the most common endophyte of the tree *Pseudotsuga menziesii* is the fungus *Rhabodocline parkii*, and it resides without causing disease directly among the epidermal cells of the leaves, through haustoria-mitigated invasion. Nitrogen-fixing *Gluconacetobacter* bacteria are abundant as endophytes in apoplastic spaces of ground (non-vascular) tissue in sugar cane. Clavicipitaceous fungal endophytes insert themselves into intercellular spaces of leaf mesophyll, and are not localized to vascular bundles. Etc., etc. So the argument that there is no possibility for endophytic microbes in moss caulids because they lack vascular tissue or other specialized compartments is not supported when we examine colonization patterns of endophytes in vascular plants.

- we do agree that we did not adequately introduce the reader to the issue of possible endophytic cyanobacteria. Our explanation for our motivation is as follows, and has been added to the introduction (current lines 108-123):

> "Molecular tests for the presence of cyanobacteria in mosses typically yield positive results and often indicate high metabolic activity, even in non-*Sphagnum* mosses (Lindo and Whiteley, 2011; Warshan et al., 2016; Groß et al. 2023). Direct microscopic observation and cultivation of the symbionts are relatively rare compared to the abundance predicted by molecular data, however. Microscopic observation and cultivation of plant-symbiotic organisms can be difficult or uncertain, requiring many hours of effort and time, possibly discouraging researchers. Compounding the difficulties, *Nostoc*-like organisms and other cyanobacteria may also exist on mosses in few-to-single cell states such as hormogonia (Büdel et al. 2024) which are probably mostly missed by visual inspection, and which are difficult to render into axenic culture. We hypothesize that another factor may also contribute to the imbalance between visual observations of cyanobacterial symbionts in moss and reports of relatively high abundance and activity: some cyanobacteria may be capable of living endophytically in mosses. This is supported by numerous studies that report the presence of putative endophytic bacteria in non-Sphagnum mosses (e.g. – Tveit et al. (2020)). As moss leaflet lamina are only one cell thick in cross-section, we

hypothesize that if such endophytic colonization occurs it is within the caulids, which are multicellular in cross-section."

The following line has also been changed:

From (original line 480):
Stems ('caulids') and rhizoids of mosses are multicellular...

To (current line 614):
Stems ('caulids') and rhizoids of mosses are multicellular in cross-section...

-This adds an additional citation to our works cited section:

"Büdel B, Friedl T, Beyschlag W (eds) (2024) Biology of algae, lichens and bryophytes. Springer Spektrum, Berlin, Heidelberg"

Lines 153-154, "Surface sterilized samples were then again inspected for visible cyanobacterial colonization": I understand that several papers claim to have sterilized the surface of mosses and other plants, but seldom do they provide evidence for that. You mentioned in the introduction that cyanobacteria "appear to prefer to epiphytically inhabit the spaces in protected leaflet-axils along moss caulids ('stems'), or in incurved regions of leaflet-tissue and other protected areas" (Lines 56-58), so it should be quite hard to sterilize the surface of the moss. Plus, you've only checked for the presence of cyanobacteria, not other microbes. I understand what you were trying to achieve here, but I think you should rephrase this section to more accurately represent what you intended to do.

- We have removed references to "surface sterilization," and changed the header of this section to "Endophyte survey" (original lines 143 and 316, current lines 209 and 426). Generally, wherever the term was used, we have substituted the words "epiphyte reduced," or "epiphyte reduction." This includes original lines 145, 153, 185, 188, 317-318, 321, Figure 7 caption, and Figure 8 caption (now supp. fig. 4).

- however, we don't agree that the incurved surfaces of moss leaflet axils were difficult to access with the surface sterilization. Penetration by the first step of ultra-sound sonication to the more protected areas of the mosses might have been poor. However the sodium perchlorate and ethanol soaks, which were in combination with surfactant and which included non-trivial vertexing steps, visibly penetrated these spaces in the mosses. We show the only image we have of remnant cyanobacteria on leaves following the surface sterilization. In any case, it is the "stems" (caulids) that are of most interest here, and these are generally quite exposed.

This brings up the additional point that surface sterilization is always a gradient-type process, rather than a process that ends simply in success or failure. The definition of endophyte, and exactly which surfaces are internal or external on a plant are at many times subjective and are continuous rather than discrete. We do not wish to delve into the debates, but merely to show that the caulids of these mosses may host a physically close and protective symbiosis, one that might enable cyanobacteria to endure strong chemical attack from outside the plant.

Line 155, "*Nostoc*-like organisms": Throughout the manuscript, some times you refer to these bacteria as "*Nostoc*" and other times as "*Nostoc*-like". Considering that *Nostoc* is indeed a polyphyletic genus and that you show another proof of that in your own Results section, I think it's more appropriate to use "*Nostoc*-like" along your manuscript and to highlight the interesting underestimated diversity of the cyanobacteria in association with mosses. You could also use nostocacean instead if you prefer to refer to their family instead of the genus.

- we agree and have corrected. However, whenever we reference specific organism from others' research, we defer to the authority and refer to organism as it is referenced in the their work.

Line 160, "filamentous bacteria that were observed in sheathed syncoenobia": Sorry, but I've never heard the term syncoenobia used when referring to cyanobacteria. Are you sure that terminology is used in this context? Could you clarify what it means here?

- agree, the term syncoenobia is pretty obscure. We found it in Büdel and Friedl's *Biology of Algae, Lichens, and Bryophytes* (2024), in fig. 3.29. The term may have originated in old works by Mollenhauer. However, we used it because we were struggling to find a term to differentiate large, compound colonies of nostoc containing numerous smaller sheath-covered sub-colonies, as compared to other simpler life stages of *Nostoc*. However, it seems to be an out-of-date term, and also perhaps misused here. We have removed it. We used the term in 3 places:

From (original line 160):

"*Nostoc*-like organisms" were defined as any filamentous bacteria that were observed in sheathed syncoenobia, especially when trichomes displayed obvious intercalary hetercytes.

To (current lines 226-228):

"*Nostoc*-like organisms" were defined as any filamentous, chlorophyllous bacteria that were observed in spherical, sheathed colonies, especially when trichomes displayed obvious intercalary hetercytes.

From (original line 398):

Nostoc-like organisms exist in many other life stages besides colony (syncoebium) stages (Mollenhauer, 1986) including single-cell and single-trichome life stages.

To (current lines 526-528):

Nostoc-like organisms exist in many other life stages besides multicellular colony stages, including single-cell and single-trichome life stages (Mollenhauer, 1986).

From (original line 550):

We found epiphytic syncoebia relatively rarely in our mosses, ...

To (current lines 619):

We found large epiphytic colonies of *Nostoc*-like organisms relatively rarely in our mosses, ...

Line 161, "hetercytes": There's a typo here. It should read heterocytes.

- spelling corrected.

Line 161 and elsewhere, "BG-110": Another small detail here, but could you please make the '0' here a subscript character? Or else it reads like 110, which is inaccurate.

- it seems this was an error in the pdf conversion. Corrected.

Lines 167 and 170, "eukaryotic algae": Here too "eukaryotic" is redundant. Please just use "algae".

- see our answer to related comment above.

Lines 176-177, "As of time of writing": I don't think you need to say this here.

- deleted

Lines 177-178, "*Nostoc*-like organisms were successfully isolated and cultivated from 4 separate moss host species": But in Table 1 you show information for 5 moss species. What happened to the fifth one?

- table 1 shows all mosses from which some sort of observation was made, not necessarily successfully cultivated. We explain in Results, section "Culture of Nostoc-like organisms" that the *Nostoc*-like organism from *Pseudoamblystegium* moss failed to grow, though we have photo-documentation of abundant growth on the moss (original lines 326-328, figure 2). Our new supplementary figure 1 may also help to clarify, as it graphically summarizes the laboratory procedures and includes a list of which *Nostoc*-like organisms were included in each step. We have also added the following clarification to the caption for table 1:

From (original line 117):
Table 1. Sampled locations.

To (current lines 179-181):
Table 1. Sampled locations and moss species. In *Orthotrichum, Nostoc*-like organisms were photographed but failed to culture. Samples from University of Bayreuth were used for PCR tests for whole-community *nifH* but not for microscopy or culturing. All other mosses were subject to microscopic inspection for- and culturing of *Nostoc*-like organisms.

Line 185, "*Anabaena variabilis*": This species has been reclassified in the genus *Trichormus* by Komarek and Anagnostidis in 1989. Please see http://www.cyanodb.cz/#/Trichormus.

- We have changed all references to *Anabaena variabilis* to *Trichormus variabilis* or *T. variabilis*. The culture we used is listed as *Anabaena* by the curators of this strain (Leibniz Institute DSMZ), so we have listed this information in the first mention in the text, as a synonym (original line 185, current line 253).

Lines 185 and 188: How were these plants surface-sterilized?

- this was an additional etOH and NaOCl sterilization, but with no sonication step or tween surfactant. The following information has been added to the text (original line 189, current lines 257-261):

"In both *Brachythecium* and *Solidago* control samples, epiphyte reduction was conducted in the following way: thorough rinse with sterile water, followed by immersion for 1 minute in 97% ethanol soak and 30 seconds in 2% NaOCl, followed by additional immersion for 1 minute in 97% ethanol and a sterile water rinse."

Line 196, NifH genes: Here too, NifH refers to the protein, while *nifH* refers to the gene.

- corrected.

Lines 196-198, "NifH genes from DNA of whole-community moss+microbiome were amplified from samples of *H. splendens* collected from the University of Bayreuth Ecological Botanical Garden": How many samples? And why was this only done for these samples? What was the objective here?

This was a preliminary test. As we discuss elsewhere, the absence of direct observations of the nostocacean-moss symbiosis in Germany were confusing to us, given their documented occurrence elsewhere and given that symbiotic *Nostoc* had been detected molecularly as +/- plentiful in German forest mosses by colleagues (Groß et al 2024). We had no reason to believe our forests were exceptions. So we hypothesized that these symbiotic *Nostoc*-like organisms were escaping visual detection. This PCR test using cyanobacterial *nifH* primers was our first indication that cyanobacteria are likely present in these mosses even when they are not detected by close microscopic inspection. This also led to our hypothesis that some of these organisms may be endophytic. Additionally it was at this phase that we observed the plant homologs of *nifH*. The only samples for this test were those represented in the gels in figure 8. We agree that this is not clear from in the text. To clarify we have reworked the last paragraph of the introduction:

From (original lines 101-103):

> We utilized a microscopy- and culture-based approach for detection and description of *Nostoc*-like organisms, which we hoped would enable precise imaging and taxonomic placement of symbionts, using full-length 16S and partial *nifH* PCR amplicon sequence data.

To (current lines 142-150):

> We hypothesized that cyanobacteria are active in moss tissues even when colonies of cyanobacteria are not visible on the moss, possibly due to endophytic colonization by cyanobacteria. We tested this by PCR-amplification of cyanobacterial-specific *nifH* genes on moss tissues with no visual colonization by cyanobacteria, and via surveys with fluorescence microscopy for cyanobacterial phycobilisomes on mosses that had been subjected to a rigorous reduction of epiphytic flora. In the case of mosses which did show obvious colonization by Nostoc-like organisms, we utilized a microscopy- and culture-based approach to confirm their presence, which we hoped would also enable precise imaging and taxonomic placement of symbionts, using full-length 16S and partial *nifH* PCR amplicon sequence data.

Lines 207-210: This is a bit confusing because later you don't mention which primers you used in this reaction, but in the next paragraph you mention a PCR of the 16S rRNA gene. So do these lines refer to *nifH* amplification in the Nostoc-like colonies? Please make this clear here.

- This section has been reworked for clarity, as we mention above.

Lines 215-216: The primer sequences are conventionally written as upper case characters.

- corrected

Lines 223-225: Only the 16S rRNA gene amplicons were purified? What about the nifH amplicons?

- we have clarified (original line 223, current line 309):

> “PCR products from both *nifH* and 16S rRNA amplifications were cleaned using...”

Lines 231-233, "SQK-LSK114 kit was used in the case of 16S rRNA gene of Hylocomium-associated Nostoc, and the remaining 6 colony-PCR products were individually barcoded, multiplexed and sequenced jointly using a SQK-RBK114.24 kit": Sorry, but this is unclear to me because you're using different terms from the previous paragraphs. So Nanopore sequencing was not only done for colony PCR products like you stated in Line 227, but also for what you called "whole community moss+microbiome"? And you sequenced one Nostoc-like colony from one moss collected in each of the six locations in Table 1?

- It is indeed a confusing paragraph, with lots of different amplicons to keep track of. We have reworked the following sentence, hopefully helping the situation:

From (original lines 231-232:

"SQK-LSK114 kit was used in the case of 16S rRNA gene of *Hylocomium*-associated *Nostoc*, and the remaining 6 colony-PCR products were individually barcoded..."

To (current lines 318-324):

"SQK-LSK114 kit was used in the case of 16S rRNA gene of *Hylocomium*-associated *Nostoc* from Waldstein, which was sequenced first and separately from later cultures. PCR of *nifH* gene from *Pseudoamblystegium*-associated *Nostoc*-like organism was not successful. The remaining 6 colony-PCR products to be sequenced consisted of *nifH* and 16S rRNA amplicons from *Nostoc*-like organisms successfully cultured from *Orthotrichum*, *Zygodon*, and *Homalia* mosses, and the *nifH* amplicons from the *Hylocomium*-associated *Nostoc*-like organism from Waldstein. These were individually barcoded..."

- we made the decision not to sequence the "whole-community moss" samples because it seemed likely that the abundance of *nifH* homologs from the host chloroplasts observed in the gels were going to swamp the flowcells, and we had limited resources for this project. We now include the following additional line to clarify that whole community moss samples were not sequenced (current lines 277-281):

"nifH PCR amplicons from whole-community moss samples were subject to gel visualization but were not further sequenced, due to an apparent abundance of *nifH* homologs from host mosses (see results). *Hylocomium* samples from University of Bayreuth were not used in any further analysis in this study."

Lines 233-234, "In the first sequencing run, with only *Hylocomium*-associated *Nostoc*": Here, do you mean the two *Hylocomium* samples you had or just one of them? And if it's the latter, was it for the colony PCR or the moss+microbiome amplicons?

- we hope that the corrections for the previous comment provide clarification here, also.

Line 254, "we used the BLAST-suite tools, specifically blastn and blastx programs": What version did you use?

- we have now included the software version (current line 351). We also added a additional citation for the BLAST+ software suite:

Camacho C, Coulouris G, Avagyan V, et al (2009) BLAST+: architecture and applications. BMC Bioinformatics 10:421. https://doi.org/10.1186/1471-2105-10-421

Line 256, "nifH amino acid sequences"; Line 257, "our custom database of nifH sequences was a concatenation of NifH sequences": Please make it clearer when are you referring to DNA sequences (i.e., nifH) and protein sequences (i.e., NifH) by using their correct names. It seems like you're using nifH and NifH interchangeably and this is very confusing.

- all sequence data in this paragraph (and only in this paragraph) were in amino acid form. So we have corrected all as suggested, from *nifH* to NifH.

Line 261, "CIAlign": What version did you use?

- version information added

Line 298, "*Nostoc azollae* (NC_014248.1)": This species was also reclassified in the genus *Trichormus* by Komarek and Anagnostidis in 1989.

- We have changed both references to *Nostoc azollae* to *Trichormus azollae* or *T. azollae* in the text and in figure 10. The genome from which the 16S was taken named as "*Nostoc azollae"* on NCBI, so we have listed this information in the first mention in the text as a synonym (original lines 297, 298, current line 404).

*Results

Line 324, "Out of 5 observed Nostoc-like organisms": But you did observe only five. In Figures 2-7, you show much more than five colonies.

- We're not sure what Reviewer 1 is requesting/observing here. In our microscopic inspection of the various moss samples, we were able to observe growth by five *Nostoc*-like organisms on 5 species of moss. Our culture+16S rRNA PCR survey supports this (4 unique 16S sequences, though in one moss all culture efforts failed, those with the *O. lyellii*-associated *Nostoc*-like organism). We hope our new supplementary figure 1 will be helpful here, as it gives an overview of our laboratory pipeline, with lists of which organisms were involved in each step.

  We show images of these five *Nostoc*-like organisms in figures 2-6. Currently, each has its own numbered figure. Within each figure, there are 1-3 images of the organism. Within each of these images there are indeed multiple colonies shown, often more than 5.

  Perhaps Reviewer 1 is suggesting that because there are multiple colonies were present on each of the mosses from which we cultivated, we cannot rule out that these multiple colonies are possibly from diverse strains of *Nostoc*-like organisms, that some of these diverse strains were left behind, and that therefore our cultures don't represent the true diversity of organisms captured in our photos. This is definitely possible. We assumed that because it was a relatively rare event to find *Nostoc*-like organisms sufficiently abundant enough to see with microscopy, these were single strains that had reached high abundance. But this was an assumption.

  In any case, more clarification is needed here from reviewer 1.

Lines 328-331, "The *Pseudoamblystegium*-associated Nostoc-like organism (…) was overtaken by an unidentified, co-cultured non-filamentous cyanobacterium (…), though some 16s data from the Nostoc-like organism were recovered": In your methods, you did not list chimera checks. So how can you ensure this sequence was legitimate?

- It is true we cannot be completely certain of the accuracy of this sequence, but it appears to lack chimeras. It aligns well with several references sequences, and varies from these reference sequences in a localized manner, in terms of one or two base pair differences throughout the length of the read. To our knowledge most of the automated chimera-checking algorithms available are intended for high-throughput sequencing datasets and rely on internal patterns of reads to detect likely chimeras, which isn't applicable here. Since we had only a handful of final consensus 16S and *nifH* sequences, this allowed us to conduct the process of hand curation that we mention in the paper: we ran small alignments with well known 16S reference sequences and blastn checks against public databases, as mentioned in the text. Readers can judge for themselves, as all consensus sequences are publicly available. We realized we didn't include accession numbers in our submission, so information about GenBank accession numbers have been added to the text. We have added the following to the bioinformatics methods section and Declarations to clarify:

From (original lines 233-235):

"In the first sequencing run, with only *Hylocomium*-associated *Nostoc*, sequencing ran until a depth of ~100,000 reads, with simultaneous fast-model basecalling using Guppy..."

To (current lines 325-327):

"In the first sequencing run, which included only 16s rRNA amplicons from the *Hylocomium*-associated *Nostoc*-like organism, sequencing ran until a depth of ~100,000 reads, with simultaneous fast-model basecalling using Guppy..."

Inserted at original line 248 (current lines 341-344):

"In the case of the sequence library that was basecalled with Dorado, the Dorado command executes includes a default chimera-check/read-splitting step for chimeric reads resulting from Nanopore sequencing error. In either case, PCR-chimeras were screened manually in the alignments discussed below."

Inserted at original line 264 (current lines 362-364):

"At this stage, further screening for chimeric reads was conducted: over-length reads were discarded, and remaining possible chimeras were discovered via hand-curation of the alignments."

Inserted at original line 533 (current lines 661-664):

"16s rRNA and *nifH* consensus sequences are available at the NCBI Nucleotide Database. GenBank accession numbers for 16s rRNA consensus sequences are: PX869778, PX856665, PX856666, PX856664. GenBank accession numbers for *nifH* consensus sequences are PX892065, PX892066 - PX892068, PX892069 - PX892071."

Line 331, "some 16s data": Please avoid using informal language in a scientific paper such as this one. This should be corrected to "some 16S rRNA gene sequence data".

- corrected (current line 443).

Lines 334-335, "Culture-free amplification of cyanobacterial nifH from Hylocomium moss tissue yielded two size classes of non-chimeric amplicons": Did you do any chimera check for these sequences? It's not listed in the Methods section. Furthermore, as far as I know, there's no way to tell if gel bands belong to chimeric sequences or not.

- as Reviewer 1 notes next, these amplicons were not sequenced. As per Reviewer 1 requests, this section has been substantially modified, see next. It would take a very consistent pattern of PCR chimera formation to yield the same bright banding pattern across 11 PCR reactions (+ controls). But as Reviewer 1 notes, we do not know for certain. The following has been changed:

From (original lines 334-336):

"Culture-free amplification of cyanobacterial nifH from Hylocomium moss tissue yielded two size classes of non-chimeric amplicons, one at the expected size of slightly less than 400 bp, and another with a size of >500 bp."

To (current lines 446-456):

Culture-free amplification of *nifH* using cyanobacterial *nifH* primers from *Hylocomium* moss tissue consistently showed amplification from moss tissues where no cyanobacterial colonization was visually detected (supp. fig. 4). Two bands in the size range of interest in the whole-community (moss + microbiome) samples were consistently observed. One band matched the expected size of nifH and matched bands presumed to *nifH* amplicons observed from *T. variabilis*-only controls, and appeared in all moss samples and sections at varying brightnesses. Another slightly larger size class band was displayed which appeared only when a host plant was present and not in amplicons from axenic culture of *T. variabilis*, possibly consisting of hosts plant homologs to *nifH*. Other, much larger amplicons were also observed, likely due to non-optimized PCR conditions, probably consisting of either non-specific amplification or concatomeric PCR products.

Lines 333-341: So the samples that you called moss+microbiomes were not sequenced, but merely PCR amplified? I'm sorry, but I really don't believe this experiment is useful for your study. Firstly, PCR of the nifH gene is notoriously difficult due to the high probability of non-specific amplification of homologous genes involved in photosynthetic processes like bchL/bchX/chlL, especially in plant-associated communities such as the ones you studied. Therefore, such work requires heavy filtering of sequencing data to select the target sequences. Secondly, determining amplicon size based on comparisons of gel bands to molecular ladders is highly inaccurate and should never be used to identify specific target products with so many bands. Thirdly, you chose the most studied (and therefore, less interesting) moss species among the ones you sampled for this experiment. It shouldn't have been difficult to do a similar procedure for the other, more uncommon mosses, and this would've been much more exciting. Lastly, there's no replication, so you can't even say much about how representative this result is. It's not clear to me what you intended with this, but regardless of that I strongly advise you to remove this part from the article.

- There is no need to be sorry. We are aware of the difficulties with this graphic and this small experiment, and its "low-tech" nature. We are aware of the difficulties of homologous genes in host plants to *nifH* and the problems this causes in plant microbiome studies. We referenced the some of the standard literature concerning these difficulties in the text (e.g. – Angel et al. 2018, original line 339, current lines 279). We agree that we did not explain our reasoning behind it very well, and that it is currently overemphasized in the manuscript, including an overemphasis on fragment size numbers. However, we respectfully disagree that it is of no value to the manuscript. We believe it is a small but adequately designed experiment, with intriguing results that support one of our hypotheses. As such we have moved the gel image to supplemental figure 4, and attempted to clarify discussion of this experiment.

Inserted at original line 193 (current line 265):

To test for the presence of cryptic colonization by nitrogen-fixing cyanobacteria in moss tissues, ...

From (original lines 417-423):

"Culture-free amplification of cyanobacterial *nifH* from *Hylocomium* moss tissue yielded two size classes of non-chimeric amplicons, one at the expected size of slightly less than 400 bp, and another with a size of >500 bp. (Fig. 8). Based on PCR controls from presumed *Nostoc*-free plant tissues and from our *Anabaena* culture, we presume that the smaller size class represents the intended target, cyanobacterial *nifH*, and the other represents non-target homologous genes from host moss tissue (Angel et al., 2018). Both amplicon types were observed in all portions of *Hylocomium* moss tissue (low, middle, upper), indicating possible cyanobacterial colonization of moss tissues even when not visible through standard microscopic inspection."

To (current lines 446-456):

Culture-free amplification of *nifH* using cyanobacterial *nifH* primers from Hylocomium moss tissue consistently showed amplification from moss tissues where no cyanobacterial colonization was visually detected (supp. fig. 4). Two bands in the size range of interest in the whole-community (moss + microbiome) samples were consistently observed. One band matched the expected size of *nifH* and matched bands presumed to *nifH* amplicons observed from T. variabilis-only controls, and appeared in all moss samples and sections at varying brightnesses. Another slightly larger size class band was displayed which appeared only when a host plant was present and not in amplicons from axenic culture of T. variabilis, possibly consisting of hosts plant homologs to *nifH*. Other, much larger amplicons were also observed, likely due to non-optimized PCR conditions, probably consisting of either non-specific amplification or concatomeric PCR products.

- In-text citation of Angel et al. (2018) has been moved to methods, current line 279.

Further justification/discussion:

-Early in our surveys we found cyanobacterial colonies via microscopic inspection only occasionally. To check if we were overlooking them, we amplified cyanobacterial *nifH* genes on some sections of the moss (caulids+phyllids) which we had thoroughly inspected with standard light microscopy for *Nostoc* and other cyanobacteria and on which we found none. There is replication (3 gametophytes × 3 heights), though this is obviously not as much as one would like. We designed positive controls for (1) actual *nifH* amplicons from PCR on DNA extracted from axenic culture of *Trichormus*/*Anabaena variabilis*, and for (2) plant *nifH* homologs using amplicons from PCR on DNA extracted from a surface-sterilized higher plant (*Solidago* sp.) and a moss (*Brachythecium* sp.) that was predicted not to have N-fixing cyanobacteria because of high-nitrate (roadside) conditions.

-Reviewer 1's objections to the use of gels for determining size is very valid but is perhaps missing the point of this small experiment. We did not actually mean to quantify exact sizes from the gel, we intended to report patterns of banding. A review of the gel shows two consistent bands in the size range of interest in the moss + microbiome samples, one of which matches the *nifH* from *T.*

variabilis and one band of a slightly large size class which appears only when a host plant is present, indicating that these amplicons could be due to host plant *nifH* homologs.

-We stopped at the gel stage upon seeing the abundance of amplicons from possible homologous genes. We did not have the appropriate sequencing depth available to filter the resulting sequences bioinformatically as advocated by Angel et al. (2018), given the limited resources for the project and the abundance of host reads. We then pursued the culture-based approach.

-The request to conduct a similar trial on other mosses is interesting, and this can perhaps be included in future efforts. However, *Hylocomium* was selected precisely because it is thought to be a common host for these symbioses, even as photo-documentation of the symbioses in the literature is actually relatively rare, and unknown from our region as far as we are aware.

Lines 350-352, "In *Zygodon*- and *Homalia*-associated cultures each appeared to host 3 alternative nitrogenases, each falling into a separate clade on the Kubota-Ininbergs *nifH* tree": You assume that you were able to get single strains via pipetting, but it's still possible that you had actually collected a few different ones belonging to different nostocacean cyanobacteria colonizing the same moss, isn't it? Unless you do genome sequencing, it's not yet possible to determine this. Correct if I'm wrong, but I don't remember seeing many reports of cyanobacteria with three alternative nitrogenases in literature.

- it is possible that we collected multiple members of Nostacaceae from the same moss samples, but there are reasons to believe we succeeded in creating largely monocyanobacterial cultures, with the exception of our attempted culture of *Nostoc*-like culture from *Pseudoamblystegium*. Each final liquid culture originated from a single, isolated colony of a *Nostoc*-like organism, and were microscopically examined checked before sequencing for *Nostoc*-like morphology and uniformity. It was in this way that we noticed the overgrowth by a non-*Nostoc*-like cyanobacteria in our *Pseudoamblystegium*-associated culture. We acknowledge that Reviewer 1 is correct, in that colonies of *Nostoc*-like organisms frequently contain other cyanobacteria or multiple *Nostoc* strains within them, but we didn't observe additional (>1) nostacacean 16s sequences in our colony PCRs, i.e. – each culture yielded only one 16s sequence belonging to Nostacaceae. An additional, minor point: in both cases where multiple *nifH* versions were observed, the read abundances from the 3 versions appeared approximately equal original (see supplemental figure 1, now supplemental figure 3), which we would expect if the reads all originated from a single organism in culture.

- It is true that *Nostoc* genomes with 3 alternative *nifH* versions do not seem to have been directly reported in the literature to date, but we came to our conclusion after finding that one of the most intensively studied *Nostoc* genomes (*Nostoc punctiforme* PCC 73102, NCBI Reference Sequence: NC_010628.1) is annotated with 3 *nifH* versions. All three of these reference *nifH* sequences are also included in *nifH* trees in both of the two previous papers that are most pertinent to our work, Ininbergs (2011) and Kubota (2023). We did not originally mention this in the text, but these three *nifH* from this PCC73102 genome were included in two of our alignments/trees of *nifH* raw reads from the nanopore sequencer. They are visible in the tree labeled "Zygodon-associated *Compactonostoc nifH* reads" in original supplemental figure 1 (currently supplemental figure 3) as the yellow branches. These alternate *nifH* genes from *N. punctiforme* are also incorporated in the tree of *nifH* reads from the *Homalia*-associated *Nostoc*-like organism, though they are not highlighted. In both, cases they aligned +/- well into the clusters of *nifH* reads from this *Nostoc*-like organism. These reference *nifH* sequences were not included in our tree of *nifH* from the *Hylocomium*-associated *Nostoc*-like organism, as in this case we had insufficient high-quality reads to test the hypothesis of multiple *nifH* versions. We will highlight these branches in the first two trees with a more visible color.

- Additionally, this comment by Reviewer 1 motivated us to quickly check the number of *nifH* genes in some (~10) other annotated Nostocaceae genomes from JGI. We found a variable number of annotated *nifH* genes in these genomes, but genomes with 2 alternate *nifH* genes were quite common, and genomes with 3-4 copies are also published, e.g. – *Nostoc* sp. 232 (*Peltigera membranaceae* symbiont, ref assembly GCF_002245985) and *Trichormus azollae* 0708 (ref seq NC_014248) both are reported to host 3-4 copies of *nifH*. We cannot vouch for the quality of these annotations without extensive further work, but taken together with the published multiple versions of *nifH* genes in *N. punctiforme* genome that we incorporated into our analysis, they give further reason to think that our observations are at least within the realm of possibility.

- given the above, we have increased the visibility of the alternate *nifH* versions from the published genome in our alignments via the current supplemental figure 3 (originally supplemental figure 1), and added the following language:

Inserted at original line 268 (current lines 370-374):

"Three reference sequences of *nifH* were extracted from the genome of *Nostoc punctiforme* PCC 73102 / ATCC 29133 (NCBI Reference Sequence NC_010628.1) and included in this alignment. These reference *nifH* genes are annotated/positioned in the genome as follows: WP_012407220.1 (position 512909-513802), WP_012407784.1 (position 1229540-1230433), WP_012412139.1 (position 7297077-7298042)."

From (original lines 350-352):

"In *Zygodon*- and *Homalia*-associated cultures each appeared to host 3 alternative nitrogenases, each falling into a separate clade on the Kubota-Ininbergs *nifH* tree."

To (current lines 466-471):

"*Zygodon*- and *Homalia*-associated cultures each appeared to host 3 alternative nitrogenases, several of which showed homology respectively with alternate *nifH* sequences extracted from the reference genome of *Nostoc punctiforme* PCC 73102 / ATCC 29133 (supp. fig. 5). In each case, these observed alternate *nifH* versions were placed into separate clades on the Kubota-Ininbergs *nifH* tree, even though originating from a single colony, and likely from single strain of *Nostoc*-like organism."

Figure 9: You placed references instead of species names in several branches of the tree. That's not so informative. Could you please provide the species instead? Also, the writing in the figure is very small and hard to see. Is it possible to increase it?

- We have adjusted the font sizes and spacing of this figure to make it more readable. The branches which are labeled only as references in figure 9 are those that originated from the works of Kubota et al. (2023) and Ininbergs et al. (2011). We would prefer to continue to use the labels as they are, which are as given by the original authors. Explanation: we do not feel confident adding taxonomic identifications on the unlabeled *nifH* sequences in this tree because identifications were never given. Due to their culture-free methods, the authors were in both papers not able to really link the *nifH* sequences they observed with 16S sequences or colony morphology. They are thought by the authors to be members of Nostacaceae based on their sequence similarity and placement in a large tree of reference Nostacaceae *nifH* sequences. They are presumed originate from either *Nostoc* or *Stigonema* by Ininberg et al. and Kubota et al. However, we are not as certain, because of our observation that a single culture of a (probably) single strain of *Nostoc*-like organism yielded diverse *nifH* versions that fall into multiple clades as discussed elsewhere in the text and in this document.

Figure 9, "nifH gene phylogeny of select members of Nostoceae": In the Methods, you mentioned you used amino acid sequences, not gene sequences for this tree. So which information is the correct one?

- Our data was always in the form of DNA sequences. It was never translated, and we never sequenced amino acids. Amino acid sequence data were used in our study only in an early step of our bioinformatic pipeline, as part of the filtering the raw nanopore sequence data. At this step we filtered raw nanopore DNA sequences of interest by retaining only close blastx matches from our raw reads to sequences from 3 published NifH amino acid databases. Blastx as opposed to blastn used was due to the formatting of the available nifH reference databases, consisted of amino acid sequences. We have added the following to clarify (including additional modifications listed above):

From (original lines 253-256):

"Therefore, as a first filtering step before full alignments, we used the BLAST-suite tools, specifically blastn and blastx programs to find high quality matches to Nostoc 16S sequences from the Silva 138.2 database (see link above), and to a custom database of nifH amino acid sequences."

To (current lines 350-355):

"Therefore, as a first filtering step before full alignments, we used the BLAST+ suite tools, version 2.9.0 (Comacho et al., 2009), specifically blastn to retain only high quality matches to nostocacean 16S sequences from the Silva 138.2 database (see link above), and blastx to retain high quality matches of our *nifH* DNA sequences to sequences from a custom database of NifH amino acid sequences, to remove possible *nifH* homologs. "

Table 2: Some extra information about how similar these sequences are would be welcome here. You could add sequence identity and coverage based on BLAST alignments, for example.

- we have added the requested information to this table. We also made two additional corrections:

- we dropped one of the two matches to the *Pseudoambylostegium*-associated *Nostoc*-like organism (KF142669), as it really wasn't clearly closer on the 16S phylogeny than several other matches, and we thought in retrospect that it just confuses the reader.

- in the process we discovered that the description of two sequences (*Compactonostoc*-like organisms KU219709 and MH598843) in this table were accidentally switched. This has been corrected.

- so the table has been changed from (line 375):

Moss host	Sister taxa / nearest neighbor
Hylocomium splendens	JN847344 *Leptogium*-associated *Nostoc* sp. (macrolichen symbiont)
Homalia trichomanoides	KM019941 *Nostoc pruniforme* SAG 62.79 (freshwater, free-living)
Amblystegium *subtile*	1. KF142669 *Gunnera*-associated *Nostoc* symbiont (vascular plant symbiont) 2. AB275347.1 Unidentified cyanobacterium Ni1-C1 (unidentified-macrolichen symbiont)

Zygodon sp.	1. MH598843 *Nostoc* sp., unpublished, classified as likely *Compactonostoc* by Cyanoseq 2. KU219709 *Compactonostoc shennongjiaensis* (Free-living on rocks)

- to (current line 498):

Nostoc-like orgnamism Sequence ID (Moss host)	Nearest neighbor (sequence ID and description)	Blastn alignment scores to nearest neighbor (identities, gaps)
PX869778 (*Hylocomium*)	JN847344 *Leptogium*-associated *Nostoc* sp. (macrolichen symbiont)	1395/1405 (99%), 8 (1%)
PX856665 (*Homalia*)	KM019941 *Nostoc pruniforme* SAG 62.79 (freshwater, free-living)	1353/1383 (98%), 6 (0%)
PX856664 (*Pseudoambylostegium*)	AB275347 Unidentified cyanobacterium Ni1-C1 (unidentified-macrolichen symbiont), classified as likely *Nostoc* by Cyanoseq	677/710 (95%), 11 (2%)
PX856666 (*Zygodon*)	MH598843 *Compactonostoc shennongjiaensis* (Free-living on rocks)	1150/1168 (98%), 8 (1%)
PX869778 (*Hylocomium*)	KU219709 "*Nostoc*" sp., unpublished, classified as likely Compactonostoc by Cyanoseq	904/915 (99%), 6 (1%)

Figure 10: This figure is even harder to read than the previous one. Can you try improving it? Also, you forgot to mention what the numbers in the nodes represent (bootstrap, perhaps?).

- To make the figure more readable, we have increased the font size of the labels, increased the opacity of the label boxes, and removed a few of the reference sequences labels, to reduce complexity of the figure a bit. As the figure is already most of a page, rather than enlarging the figure, we would prefer to offer a link in the manuscript to a larger figure, should the reader like to make a closer inspection.

- there is some confusion here, as there are no branch support numbers included in this figure. This was intentional as, unlike our *nifH* phylogenies, we did not intend to provide a highly detailed 16S-based phylogeny of Nostocaceae. We would prefer to leave this to the relevant cyanobacterial experts. The goals of this figure were (1) to provide a cladogram showing the general trend that no single clade of moss symbionts emerges from the 16S data that we have (although genetic distances are hypothesized by the tree building algorithm). Instead, the symbionts we observed seem to be from all over the family of Nostacaeae. The second goal (2) for this tree was to identify nearest neighbors of our cultured organisms, as we felt this would be more informative for taxonomic identification of these *Nostoc*-like organisms than simple blast matches, since SSH-ALIGN + FastTree pipeline we used takes secondary structure of 16S sequences into account. Branch support measures and scale bar were not included in this figure, because we felt they would crowd an already very busy figure with unnecessary information. However, if reviewer 1 feels this would be appropriate we can add this information back into the figure, or alternatively create a supplementary figure with the full information as we did with our *nifH* tree.

- as such, we interpret that Reviewer 1 is asking for us to explain support values from our *nifH* phylogenetic trees (previous figure 9 and supplemental figure 2). We have inserted an explanation of the confidence measures and branch lengths to the figure captions of figure 9 and supplemental figure 5:

> "Branch lengths are proportionate to expected number of nucleotide substitutions per site. Branch support values are Shimodaira-Hasegawa (SH) test values, ranging from 0 (low confidence) to 1 (high-confidence)."

*Discussion

Lines 389-391, "Given the documented importance of moss-hosted cyanobacterial nitrogen in other systems (discussed above), the contribution of these symbioses to the primary productivity of montane forest ecosystems of our region of Central Europe is also likely to be significant": This would depend on basically three factors: 1) How strongly colonized by nitrogen-fixing cyanobacteria these mosses are; 2) How much nitrogen fixation these cyanobacteria could potentially carry out; and 3) How widespread are these mosses. I don't think you can say anything about the first two since you didn't quantify cyanobacterial colonization or measure nitrogen fixation, but I assume there are surveys in the region that point out how common these mosses are or maybe even estimate their biomass is. Could you add some information about this?

We are not aware of any data sets that quantify surface coverage or biomass of these mosses on a landscape scale for our region. Their rarity/commonness can be estimated by observation data collated by the Bavarian Lichen and Moss society at https://bayern.moose-deutschland.de. We have now included maps from this website as supplemental figure 6, showing for instance that *Hylocomium* is observed everywhere in the region of Bavaria.

Line 396, "mono-algal": You meant mono-cyanobacterial, right?

- corrected.

Lines 403-404, "As such, the organisms cultured here likely greatly under-represents the diversity of *Nostoc*-like organisms in mosses in our region": That's possible, but it's also possible that nitrogen-fixing cyanobacteria from other families are present. You should not restrict it to nostocacean cyanobacteria.

- agree. We have added the following:

From (original lines 403-404):
"As such, the organisms cultured here likely greatly under-represents the diversity of *Nostoc*-like organisms in mosses in our region."

To (current lines 532-534):
"As such, the organisms cultured here likely greatly under-represents the diversity of *Nostoc*-like organisms in mosses in our region, and even more so the diversity of nitrogen-fixing cyanobacteria and nitrogen fixers generally in these mosses."

Lines 429-431: Do you have a reference for that information?

- citations have been added for the distribution of *Orthotrichum* and *Zygodon* at current lines 563, and for gametophyte sizes and the production of gemmae at current lines 564 and 568. The micro-habitat preferences of *Zygodon* spp. is from numerous personal observations of these mosses during diversity and conservation surveys in our region. The following two sources have been added:

Frahm J-P, Frey W (2004) Moosflora, 4th edn. E. Ulmer

Meinunger L, Schröder W, Dürhammer O (2007) Verbreitungsatlas der moose Deutschlands. Regensburgische Botanische Gesellschaft von 1790 e. V. (https://bayern.moose-deutschland.de)

Lines 470-471, "the Nostoc strain cultured from Homalia trichomanoides, closely related to Nostoc pruniforme SAG 62.79 (Fig. 10), has a version of nifH that falls within this "Stigonema" cluster": Like I said above, it's possible that Stigonema cells were present in the sequenced sample?

-We have given our response to these issues in detail, see above. Microscopic inspection of cultures prior to DNA extraction, accompanying single 16S sequences from cultures, alignments of reference alternate *nifH* sequences from a well-known *Nostoc* genome, and the +/- even abundances of *nifH* read abundances from all lead us to believe that we achieved mono-cyanobacterial cultures, except where noted (*Pseudoamblystegium*-associated culture). *Stigonema* spp. have a very distinct morphology and if they were present in sufficient abundance to provide ~1/3 of all *nifH* reads, we probably would have noted them microscopically in our cultures, and/or found their 16S sequences in our raw reads.

Line 474, "We also observed the possible endophytic colonization of moss stems by cyanobacteria": That's quite unlikely, considering how anatomically simple moss stems are. An insufficient surface sterilization procedure is a more likely explanation for what's seen in the microscopy pictures. I think you need to discuss this from a moss structural perspective, as well as providing additional evidence for it.

-Our detailed response is given above. We do not agree that endophytes require structural complexity to exist within plant tissues, examples are given above. Apoplastic spaces in plants are sufficient for endophytic colonization, even for larger fungal endophytes and in ground tissue. As mentioned, the definition of "endophyte" and the purpose of the term are debatable and exist on a continuum, and here we simply wish to show that a very tight, structurally protective symbiosis may be occurring in/on the stems of these mosses. We have no reason to believe that this surface sterilization protocol was insufficient for this purpose. Indeed, we were unable to use a more rigorous protocol – during protocol development we tested slightly longer (+1 min) exposure to sodium hypochlorite and plant tissues began to discolor, indicating that intracellular structures were being degraded.

Reviewer 2

We thank reviewer 2 for their useful feedback and their encouraging words. Our responses are as follows:

Title: The title is very long and not particularly catchy.

- Noted. We propose instead:

"Expanding the geographic and host range of moss-Nostocacean symbioses in central Europe"

*Abstract

L 6-8: Can you provide a rationale for this study - was it conducted solely out of interest in the symbiosis between Nostoc and mosses?

- it was indeed. The symbiosis wasn't documented directly yet from our region, so we went looking for it. We had unique opportunity to do so due to one author's ongoing moss diversity surveys in the region, but explaining this situation seems outside the scope of the paper. However, to clarify the motivations and goals of the study, we added the following list of goals, in the form of questions, to the end of the introduction (current lines 152-163):

“”“

We ask the following questions:

- Are nitrogen fixing cyanobacteria growing in symbioses with feather mosses such as *Hylocomium splendens* in the mountain ranges of central Europe, just as they are commonly observed to do in boreal forests?
- Are there nitrogen fixing cyanobacteria on mosses from other groups beyond feather mosses, such as the families Orthotrichaceae (*Orthotrichum* and *Zygodon* spp.) and Neckeraceae (*Homalia trichomanes*)?
- Are moss-cyanobacterial symbioses present on tree-epiphytic mosses in central European forests?
- Which species of *Nostoc* and *Nostoc*-like organisms are present in these mosses, and what are their evolutionary relationships within Nostacaceae?
- Given the predicted abundances of these symbioses, where are symbiotic cyanobacteria located on their host mosses? Are some cyanobacteria present as endophytes?

“””

L 11-12: It should be noted that the mosses were sampled from the forest canopy (see title). What trees species?

- most of the host mosses (4 out 5 moss species from three tree species) were from the canopy. We have added this information concerning host trees to the abstract.

L17: What does 'single Nostoc-like organisms' mean? Does this refer to non-Nostoc diazotrophs? This should be clarified.

- agree this is unclear. Even though we use the term “*Nostoc*-like organism” frequently in the text (especially after the above edits), we do not wish to overburden the abstract with an explanation of this term. So in the abstract we have changed “*Nostoc*-like organism” to “*Nostoc* and *Nostoc*-related organism". For the main text we provide an immediate explanation of the term “*Nostoc*-like organism” prior to further use in the introduction, current line 36 (in addition to the definition provided at original line 160, current lines 226-228):

> “Researchers throughout the twentieth century continued to note frequent associations of Nostoc and closely related organisms (hereafter referred together as “Nostoc-like organisms”)...”

The following changes have also been made, hopefully this clarifies:

From (original lines 16-17):
“We note the presence of multiple alternate nifH genes within the genomes of single *Nostoc*-like organisms.”

To (current lines 22-23):

“We note the presence of multiple alternate *nifH* genes within the genomes of our cultured *Nostoc* and *Nostoc*-related organisms.”

From (original lines 23-28):

“Researchers throughout the twentieth century continued to note frequent associations of *Nostoc*-like organisms and mosses...”

To (current lines 34-36):

“Researchers throughout the twentieth century continued to note frequent associations of *Nostoc* and closely related organisms (hereafter referred together as “*Nostoc*-like organisms”) and mosses...”

L 66-67: the sentence should be rephrased: '…. involving a transfer of fixed organic nitrogen in the form of fixed, organic nitrogen from heterocystous Nostoc colonies to the host plant…'

- typo. Corrected, see reviewer 1 comments above.

L93-95: This sentence could be deleted. Specific sampling regions and locations could be given in Material and Methods.

- deleted

L100-101: Because of atmospheric deposition, nitrogen availability has been relatively high in German forests for many decades, which has often led to eutrophication and other negative effects. The question is rather whether nitrogen fixation occurs despite the high N deposition rate.

- high nitrogen fixation in forest mosses has been documented in our region despite high atmospheric deposition of N by Groß et al. (2024). However, we agree that this sentence doesn’t make sense as is, and our hypothesis needs clarification. We have added more language to clarify:

From (original lines 99-101):

“We hypothesized them as likely candidates to host the symbiosis, given the presumed low nitrogen availability and dynamic microclimate of the canopy setting.”

To (current lines 130-141):

“We hypothesized them as likely candidates to host the symbiosis, as we hypothesize that biologically available nitrogen abundances are highly dynamic in tree canopies. Our reasoning is as follows: atmospheric nitrogen deposition in central Europe is quite high (Lüers et al., 2017), but despite this biological nitrogen fixation by moss microbiomes has been observed (Groß et al., 2024). We hypothesized therefore that abundance of biologically available nitrogen in the micro-habitat of forest mosses — and especially epiphytic mosses — is highly dynamic, following a boom-or-bust pattern. At times nitrogen is probably abundant, such as during rain events when throughfall and stemflow leaches ammonium salts and DON downward and through moss tissues (Salemaa et al., 2020). At other, drier periods when the main source of N is limited to atmospheric deposition, these relatively small mosses are small “targets” to receive such stochastic deposition, and thus may benefit from the presence of N-fixing microbes.”

- This adds two sources to our reference list:

Lüers J, Grasse B, Wrzesinsky T, Foken T (2017) Climate, air pollutants, and wet deposition. In: Energy and Matter Fluxes of a Spruce Forest Ecosystem. Springer, pp 41–72

Salemaa M, Kieloaho A-J, Lindroos A-J, et al (2020) Forest mosses sensitively indicate nitrogen deposition in boreal background areas. Environmental Pollution 261:114054. https://doi.org/10.1016/j.envpol.2020.114054

*Discussion

L385-387: Reference is needed.

- reference for the distribution of the mosses we sampled are now supplied as supp. fig. 6, along with the backing reference for the source website (Meinunger et al.).

L389-393: For such statements, direct measurements of nitrogen fixation and moss biomass in the forests would be necessary. It can be assumed that mosses play only a minor role in supplying nitrogen to forests in Central Europe. However, symbiosis with diazotrophs is likely an important factor in the mosses' own nitrogen supply.

- We are suggesting that this common assumption is now worth critical investigation, given our results and the results of Groß et al. (2024) from mosses in german forests. They observed relatively high rates of nitrogen fixation, with a grand mean of perhaps ~20 micrograms per gram dry moss per day, and with values ranging from 0-95 µg N g^{-1} day^{-1}. These values are quite in-line with observed nitrogen fixation rates from boreal sites, e.g. Stuiver et al. (2014). To make our case here we could do a quick, back-of-napkin projection, using the nitrogen fixation rates found by Groß et al., Stuiver's estimates of dry moss mass / area, and an estimate of 50% moss ground cover in German coniferous forests by Diekmann et al. (2023):

(20 µg N / 1 g dry moss / 1 day) * (1 g N / 10^6 µg N) * (618 g dry moss / 1 m^2) *
(10^4 m^2 / 1 ha) * (0.5 moss ground cover) * (365 days/year) * (1 kg / 1000 g)
= 22.55 kg N / ha / year fixed by bacteria in mosses.

Obviously actual natural nitrogen fixation rates by mosses in German forests probably won't reach this number, for multiple reasons. It is interesting, however, to note that this projected rate of nitrogen fixation by moss microbiomes is in the same order of magnitude as rates of the bulk atmospheric N deposition onto most German forests. Thus it seems worth retaining some language around the larger scale ecological importance to the forest, not just to the host mosses themselves.

-As reviewer 2 notes, we may have a gotten a bit excited in our description of the ecological importance given the limited scale of our study, so we have made the following change:

From (original lines 389-93):

"Given the documented importance of moss-hosted cyanobacterial nitrogen in other systems (discussed above), the contribution of these symbioses to the primary productivity of montane forest ecosystems of our region of Central Europe is also likely to be significant. This argues for the protection of forest-floor moss layers and canopy mosses in the montane forests of the region as part of maintenance of these forest stands."

To (current lines 516-521):

"The contribution of these symbioses to the overall nitrogen input into forest ecosystems of our region of Central Europe may be significant, given the high rates of nitrogen fixation by moss-microbiomes in observed in German forests (Groß et al. (2024), the documented importance of moss-hosted cyanobacterial nitrogen in boreal forests and other ecosystems (discussed above), and the abundance of these mosses in our region. "

L434-435: Reference is needed.

- added.

L453-460: These statements are somewhat speculative, as there is no evidence of an Mo limitation in the canopy. A deficiency in other nutrients or micronutrients might play a greater role than a deficiency in Mo.

- we stated that we were speculating in the original manuscript. So we agree. References to Molybdenum have been removed:

From (original lines 453-460):

"We speculate that in the forest canopy environment, the availability of molybdenum may be limiting, requiring alternate nitrogenases with other co- factors. Alternatively, the dynamic microclimate of the forest canopy may select for homologs of nitrogenase genes with different environmental optima. Either possibility would enable diazotrophic cyanobacteria to fix nitrogen and continue growth under a wider range of conditions."

To (current lines 591-594):

"The dynamic habitat of the forest canopy may select for homologs of nitrogenase with different environmental optima and nutrient requirements, enabling diazotrophic cyanobacteria to fix nitrogen and continue growth under a wider range of conditions."

*Conclusions

The conclusion is a bit too long-winded (in particular, L. 509-520) and could be summarized more concisely.

- deleted.

L508-509: N limitation, s. comment above

- response above

Additional corrections

- *Zygodon dentatus* was in several places mis-labeled as "*Zygodon rupestris*". All references to *Zygodon* are now to *Zygodon dentatus*.

- supplemental figures have been added to the document itself, at the end of the pdf, for easier access for reviewers, since significant changes have occurred there.

- we changed the word “paralogs” on original line 497 to “versions” on current line 633.